Partial Differential Equations

Motivation for PDEs: Conservation law (1/2)

Consider a transport problem coming from the gas dynamics, traffic, population dynamics, wave equations. Focus for example on gas dynamics where we describe the motion of a gas inside a tube (1D-like to simplify), where in the transversal direction the density and the velocity of the gas is constant.

We define the density of a gas $\rho(t, x)$ such that the mass of the gas in a given section $[x_1, x_2]$ gives the mass in that section at time t by integrating the density

$$m_{[x_1,x_2]} = \int_{x_1}^{x_2} \rho(t, x)\mathrm{d}x.$$

The walls of the tube are impermeable, no exchange of mass happens across these walls, so the change of mass is happening only across the points x_1 and x_2. The rate of flow at these points is given by the velocity v times the density, i.e.,

$$\text{mass flux at } (t, x) = \rho(t, x)v(t, x).$$

Motivation for PDEs: Conservation law (2/2)

The rate of change of the mass in $[x_1, x_2]$ in a certain time step $[t_1, t_2]$ is then the difference of the flow at the two endpoints integrated in time, i.e.,

$$\int_{x_1}^{x_2} \rho(t_2, x)\,\mathrm{d}x - \int_{x_1}^{x_2} \rho(t_1, x)\,\mathrm{d}x = \int_{t_1}^{t_2} \rho(t, x_1)v(t, x_1)dt - \int_{t_1}^{t_2} \rho(t, x_2)v(t, x_2)dt.$$

This is the **integral form** of the **conservation law** (of mass).
Supposing that all the quantities are differentiable, we can differentiate in x_2 and in t_2 and obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho(t, x) + \partial_x(\rho(t, x)v(t, x)) = 0.$$

This is the differential form of the **conservation law**.

For the moment we suppose that the velocity is known a priori. More in general, we can write the conservation law as

$$\partial_t \rho(t, x) + \partial_x f(\rho(t, x)) = 0.$$

PDE Cangiani's notes

Given a domain $\Omega \in \mathbb{R}^d$ where $d > 1$, we seek for $u : \Omega \to \mathbb{R}^s$ where $s \in \mathbb{N}_0$, solution of a **stationary PDE** of order k:

$$F(x, t, u, \nabla u \ldots, \nabla^{(k-1)} u, \nabla^{(k)} u, g) = 0$$

with $g : \Omega \to \mathbb{R}^s$ a given function called **datum**. Or, more explicitely as

$$\mathcal{P}(u, g) \equiv F(x, u, \frac{\partial u}{\partial x_1}, \ldots, \frac{\partial u}{\partial x_d}, \frac{\partial^2 u}{\partial x_1 \partial x_1}, \ldots, \frac{\partial^{p_1 + \cdots + p_d} u}{\partial^{p_1} x_1 \, \partial^{p_2} x_2 \ldots \partial^{p_d} x_d}, g) = 0,$$

where $p_1 + \cdots + p_d \leq k$.

A **non stationary** PDE of order k reads: find $u : \Omega \times [0, T] \to \mathbb{R}^s$

$$\mathcal{P}(u, g) \equiv F(x, t, u, \frac{\partial u}{\partial t}, \frac{\partial u}{\partial x_1}, \ldots, \frac{\partial u}{\partial x_d}, \frac{\partial^2 u}{\partial x_1 \partial x_1}, \ldots, \frac{\partial^{p_0 + p_1 + \cdots + p_d} u}{\partial^{p_0} t \, \partial^{p_1} x_1 \ldots \partial^{p_d} x_d}, g) = 0,$$

where $p_0 + \cdots + p_d \leq k$.

A classical solution of a PDE is a function $u \in \mathcal{C}^k(\Omega \times [0, T])$ that solves the previous equation.

Definition

If the PDE can be written in the form

$$\mathcal{P}(u,g) = a(x)u + b_0(x)\partial_t u + b_1(x)\partial_{x_1} u + \cdots + b_d(x)\partial_{x_d} u + c_{\alpha(2,0,\ldots,0)}\partial_{tt} u + \cdots + \gamma_{\alpha(p_0,\ldots,p_d)} \frac{\partial^{p_0+\cdots+p_d} u}{\partial^{p_0} t\, \partial^{p_1} x_1 \ldots \partial^{p_d} x_d} + \cdots - g = 0,$$

i.e., if the coefficitents of the unknown u and of its derivatives depend only on the independent variables (t, x), then the PDE is **linear**. Else, it is **nonlinear**.

Definitions

Consider a nonlinear PDE of order k

- if the coefficients of the derivatives of order k depend only on the indepenedent variables (t, x), then the PDE is **semilinear**;
- if the coefficients of the derivates of order k depend on the independent variables (t, x) and on the partial derivatives of u of order at most $k-1$, then the PDE is **quasi-linear**;
- if it's not quasi-linear, its **fully nonlinear**.

Examples

- Reaction-advection-diffusion equation

$$\partial_t u = u_{xx} + c u_x + u^2,$$

  - is semilinear.
- Inviscid Burgers' equation

$$\partial_t u + u u_x = 0,$$

  - is quasi-linear but not semilinear.
- The Korteweg-de Vries (KdV) equation

$$\partial_t u + u \partial_x u + \partial_{xxx} u = 0,$$

  - is semilinear.
- The Monge-Ampère equation

$$u_{xx} u_{yy} - (u_{xy})^2 = 0$$

  - is fully nonlinear.

Homework

Classify the following PDE as linear, semilinear, quasi-linear or fully nonlinear:

1. Transport with time-dependent velocity: $u_t + \sin(t) u_x = 0$
2. Viscous Burgers' $u_t + u u_x + u_{xx} = 0$
3. Navier-Stokes equations

$$\begin{cases} \partial_t u + (u \cdot \nabla) u - \nu \Delta u + \nabla p = f, \\ -\mathrm{div} u = 0. \end{cases}$$

4. Eikonal equation: $|\nabla u| = 1$.
5. $\partial_t u \partial_{xx} u - u \partial_x u = 0$.

First order linear PDE , a.k.a. transport equation

$$u_t(x,t) + u_x(x,t) = 0$$

How do I found a general solution $u \in \mathcal{C}^1(\mathbb{R}^2)$?

Let's try this change of variables

$$(x,t) \to (\xi,\eta), \qquad \xi(x,t) = x+t,\ \eta(x,t) = x-t$$

with inverse

$$x(\xi,\eta) = \frac{\xi+\eta}{2},\ t(\xi,\eta) = \frac{\xi-\eta}{2}.$$

I substitute the new variables: $v(\xi,\eta) := u(x(\xi,\eta),t(\xi,\eta))$

$$\begin{aligned} u_x &= v_\xi \xi_x + v_\eta \eta_x = v_\xi + v_\eta \\ u_t &= v_\xi \xi_t + v_\eta \eta_t = v_\xi - v_\eta \end{aligned}$$

obtaining a new PDE

$$0 = u_t + u_x = 2v_\xi \iff v_\xi = 0.$$

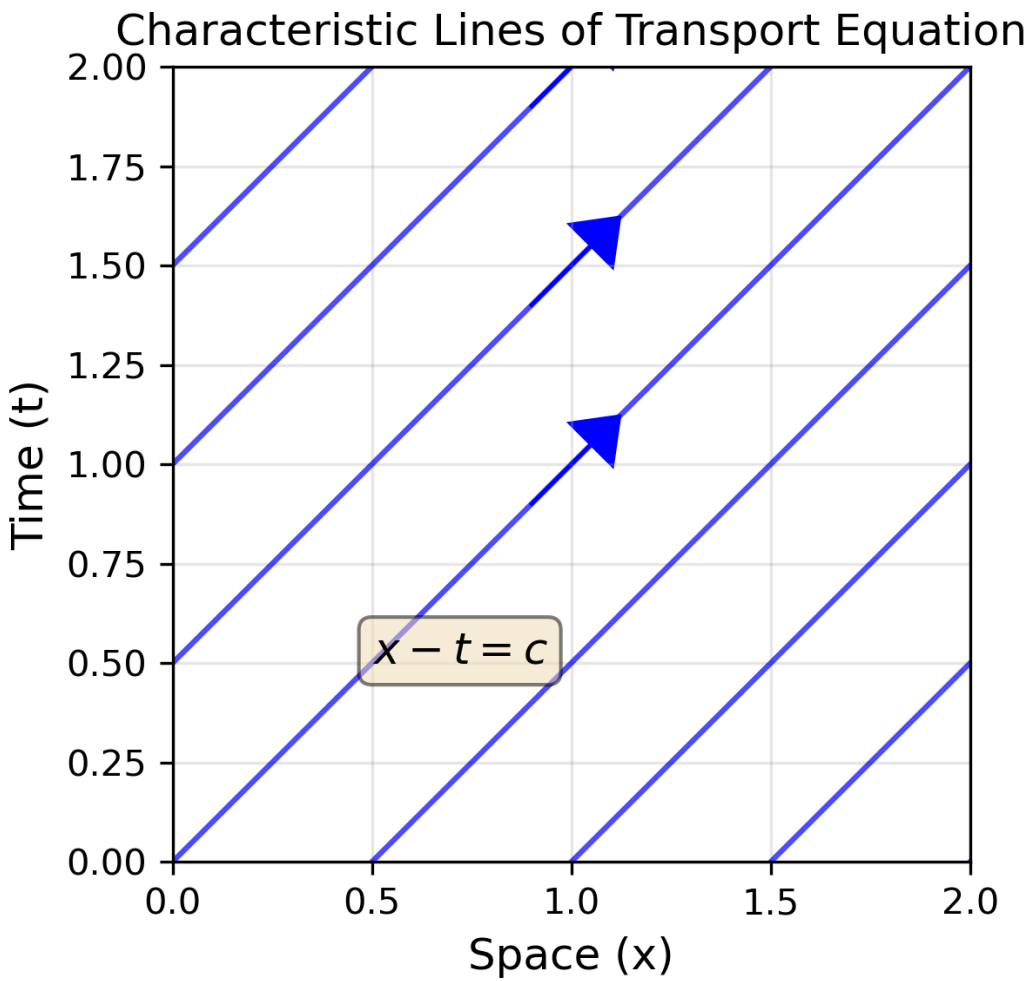
This implies that $v(\xi,\eta) = f(\eta)$ with $f \in \mathcal{C}^1(\mathbb{R})$. Going back to the original variables

$$u(x,t) = v(\xi(x,t),\eta(x,t)) = f(\xi(x,t)) = f(x-t)$$

Characteristic lines

$$u_t + u_x = 0, \qquad u(x,t) = v(\xi(x,t), \eta(x,t)) = f(\xi(x,t)) = f(x-t).$$

$$X_{x_0}(t) = x_0 + t$$



Generalization to different coefficients

$$a(t,x)u_t + b(t,x)u_x + cu(t,x) = g(t,x), (t,x) \in \Omega \subset \mathbb{R}^2.$$

Well defined (non-singular and $\mathcal{C}^1$) transformation $(t,x) \Leftrightarrow (\xi,\eta)$, i.e.,

$$\left|\frac{\partial(\xi,\eta)}{\partial(t,x)}\right| := \left|\begin{pmatrix} \xi_t & \xi_x \\ \eta_t & \eta_x \end{pmatrix}\right| = \xi_t\eta_x - \xi_x\eta_t \neq 0.$$

Change of variables: $u_t = v_\xi \xi_t + v_\eta \eta_t$, $u_x = v_\xi \xi_x + v_\eta \eta_x$, giving

$$(a\xi_t + b\xi_x)v_\xi + (a\eta_t + b\eta_x)v_\eta + cv = g(t(\xi,\eta), x(\xi,\eta))$$

Goal: simplify the previous equation, we choose η such that

$$a\eta_t + b\eta_x = 0,$$

so that we obtain an ODE for every η

$$v_\xi + \frac{c}{a\xi_t + b\xi_x} v = \frac{g(t(\xi,\eta), x(\xi,\eta))}{a\xi_t + b\xi_x}.$$

Generalization to different coefficients

To obtain $a\eta_t + b\eta_x = 0$, one should notice that, w.l.o.g., we are looking for a curve $x(t)$ such that $\eta(t, x(t)) = \eta_0$ constant for every t.

$$0 = \frac{d\eta(t, x(t))}{dt} = \eta_t + \eta_x \frac{\partial x}{\partial t} \implies \frac{\eta_t}{\eta_x} = -\partial_t x(t)$$

Hence, we have

$$\frac{\eta_t}{\eta_x} = -\frac{b}{a} \iff \partial_t x(t) = \frac{b}{a}.$$

Integrating this ODE, one obtains the curve $x(t)$, leading to the definition of $\eta(t, x)$ solving for the constant η_0.

Example

$$xu_t - tu_x = 1$$

$(x\xi_t - t\xi_x)v_\xi + (x\eta_t - t\eta_x)v_\eta = 1$

$(x\eta_t - t\eta_x) = 0$ We look for $x(t)$: $\dfrac{d\eta(t, x(t))}{dt} = \eta_t + \eta_x \dfrac{dx}{dt} = 0$

$$\frac{dx}{dt} = -\frac{t}{x} \Longrightarrow \int x\,dx = \int -t\,dt \Longrightarrow \frac{x^2}{2} = -\frac{t^2}{2} + \underbrace{C}_{\geq 0}$$

$x = \sqrt{\eta_0^2 - t^2}$

$$\eta(t, x) := \sqrt{t^2 + x^2} \qquad \Longrightarrow \qquad \eta_t = \frac{t}{\sqrt{t^2 + x^2}}, \quad \eta_x = \frac{x}{\sqrt{t^2 + x^2}},$$

$$\text{I choose } \xi(t, x) = \arctan(x/t) \qquad \Longrightarrow \qquad \xi_t = \frac{t^2}{x^2 + t^2}\frac{-x}{t^2} = -\frac{x}{x^2 + t^2}, \quad \xi_x = \frac{t}{x^2 + t^2},$$

$t = \eta\cos(\xi),\ x = \eta\sin(\xi),$

$$\underbrace{\left(-\eta\sin(\xi)\frac{\eta\sin(\xi)}{\eta^2} - \eta\cos(\xi)\frac{\eta\cos(\xi)}{\eta^2}\right)}_{=-1} v_\xi = 1, \Longrightarrow v = -\xi + f(\eta), \quad u = -\arctan(x/t) + f(x^2 + t^2).$$

- Homework: check when the transformation is invertible, draw the characteristic lines $\eta(t, x) = c$.

Homework

- Solve $u_x - 2u_y = 0$
- Solve $u_t - t^2 u_x = 1$
- Solve $yu_x - xu_y + uy = xy$ (difficult)

Second order linear PDE in 2D

Consider the PDE on $\Omega \subset \mathbb{R}^2$

$$\mathcal{P}(u, g) = A\partial_{xx}u + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu - g = 0 \quad \forall (x, y) \in \Omega$$

where $u \in \mathcal{C}^2(\Omega)$ and $A, B, C \in \mathcal{C}^2(\Omega)$ and they do not vanish simultaneously. Let's **classify** the PDE depending on the *discriminant*

$$\Delta := B^2 - 4AC.$$

Definition

- If $\Delta > 0$ the PDE is said to be **hyperbolic** (at a point (x, y))
- If $\Delta = 0$ the PDE is said to be **parabolic** (at a point (x, y))
- If $\Delta < 0$ the PDE is said to be **elliptic** (at a point (x, y))

- **Hyperbolic example: wave equation**

$$\partial_{tt}u - c\partial_{xx}u = 0 \text{ with } c > 0$$

  - Indeed, $\Delta = 4c > 0$.

- **Parabolic example: heat equation**

$$\partial_{t}u - c\partial_{xx}u = 0 \text{ with } c > 0$$

  - Indeed, $\Delta = 0$.

- **Elliptic example: Poisson equation**

$$-c\partial_{xx}u - c\partial_{yy}u = -c\Delta u = f \text{ with } c > 0$$

  - Indeed, $\Delta = -4c^2 < 0$.

- **Changing sign example: Tricomi equation**

$$yu_{xx} + u_{yy} = 0$$

  - $\Delta = -4y$.

Theorem

The sign of the discriminant Δ is invariant under smooth non-singular transformation of coordinates (i.e. under a change of variables).

Proof (not asked at exam) 1/3

We focus only on the second order terms as the first order ones do not contribute to the discriminant. Suppose we perform a smooth change of variables $(x, y) \mapsto (\xi, \eta)$, given by a diffeomorphism. Under this transformation, the second-order derivatives transform as follows:

$$u_{xx} = \alpha^2 u_{\xi\xi} + 2\alpha\beta u_{\xi\eta} + \beta^2 u_{\eta\eta},$$
$$u_{xy} = \alpha\gamma u_{\xi\xi} + (\alpha\delta + \beta\gamma) u_{\xi\eta} + \beta\delta u_{\eta\eta},$$
$$u_{yy} = \gamma^2 u_{\xi\xi} + 2\gamma\delta u_{\xi\eta} + \delta^2 u_{\eta\eta},$$

where

$$\alpha = \frac{\partial \xi}{\partial x}, \quad \beta = \frac{\partial \eta}{\partial x}, \quad \gamma = \frac{\partial \xi}{\partial y}, \quad \delta = \frac{\partial \eta}{\partial y}.$$

Proof 2/3

Rewriting the PDE in the new coordinates, we get

$$
\begin{aligned}
&Au_{xx} + Bu_{xy} + Cu_{yy} = \\
&A(\alpha^2 u_{\xi\xi} + 2\alpha\beta u_{\xi\eta} + \beta^2 u_{\eta\eta}) + B(\alpha\gamma u_{\xi\xi} + (\alpha\delta + \beta\gamma)u_{\xi\eta} + \beta\delta u_{\eta\eta}) + C(\gamma^2 u_{\xi\xi} + 2\gamma\delta u_{\xi\eta} + \delta^2 u_{\eta\eta}) = \\
&(A\alpha^2 + B\alpha\gamma + C\gamma^2)u_{\xi\xi} + (2A\alpha\beta + B(\alpha\delta + \beta\gamma) + 2C\gamma\delta)u_{\xi\eta} + (A\beta^2 + B\beta\delta + C\delta^2)u_{\eta\eta} = \\
&A' u_{\xi\xi} + B' u_{\xi\eta} + C' u_{\eta\eta}.
\end{aligned}
$$

with the transformed coefficients A', B', C' are given by

$$
\begin{aligned}
A' &= A\alpha^2 + B\alpha\gamma + C\gamma^2, \\
B' &= 2A\alpha\beta + B(\alpha\delta + \beta\gamma) + 2C\gamma\delta, \\
C' &= A\beta^2 + B\beta\delta + C\delta^2.
\end{aligned}
$$

Proof 3/3

$$
\begin{aligned}
A' &= A\alpha^2 + B\alpha\gamma + C\gamma^2, \\
B' &= 2A\alpha\beta + B(\alpha\delta + \beta\gamma) + 2C\gamma\delta, \\
C' &= A\beta^2 + B\beta\delta + C\delta^2.
\end{aligned}
$$

Now, computing the transformed discriminant:

$$
\begin{aligned}
\Delta' &= B'^2 - 4A'C' \\
&= (2A\alpha\beta + B(\alpha\delta + \beta\gamma) + 2C\gamma\delta)^2 \\
&\quad -4(A\alpha^2 + B\alpha\gamma + C\gamma^2)(A\beta^2 + B\beta\delta + C\delta^2).
\end{aligned}
$$

Expanding both terms and simplifying, we find that

$$
\begin{aligned}
\Delta' =& 4A^2\alpha^2\beta^2 + 4AB\alpha\beta(\alpha\delta + \beta\gamma) + 8AC\alpha\beta\gamma\delta + B^2(\alpha\delta + \beta\gamma)^2 + 4BC(\alpha\delta + \beta\gamma)\gamma\delta + 4C^2\gamma^2\delta^2 \\
&- 4(A^2\alpha^2\beta^2 + AB\alpha^2\beta\delta + AC\alpha^2\delta^2 + AB\alpha\beta^2\gamma + B^2\alpha\beta\gamma\delta + BC\alpha\gamma\delta^2 + AC\gamma^2\beta^2 + BC\gamma^2\beta\delta + C^2\gamma^2\delta^2) \\
=& A^2(4\alpha^2\beta^2 - 4\alpha^2\beta^2) + AB(4\alpha\beta(\alpha\delta + \beta\gamma) - 4\alpha\beta(\alpha\delta + \beta\gamma)) + AC(8\alpha\beta\gamma\delta - 4\alpha^2\delta^2 - 4\gamma^2\beta^2) \\
&+ B^2((\alpha\delta + \beta\gamma)^2 - 4\alpha\beta\gamma\delta) + BC(4(\alpha\delta + \beta\gamma)\gamma\delta - 4\gamma\delta(\alpha\delta + \gamma\beta)) + C^2(4\gamma^2\delta^2 - 4\gamma^2\delta^2) \\
=& (B^2 - 4AC)(\alpha\delta - \beta\gamma)^2 = \Delta \det(J)^2,
\end{aligned}
$$

where J is the Jacobian matrix of the transformation. Since $\det(J)^2 \geq 0$, the sign of Δ remains unchanged. This proves the invariance of the discriminant sign under a change of variables.

Hyperbolic canonical form: example wave equation

Consider the wave equation

$$\partial_{tt}u - c^2\partial_{xx}u = 0$$

with $c > 0$, so $\Delta = B^2 - 4AC = 0^2 - 4(-c) = 4c > 0$. We can find a change of variables $(x, t) \mapsto (\xi, \eta)$ such that the PDE simplifies to

$$\partial_{\xi\eta}v = 0.$$

The map is defined by

$$\eta = x + ct, \quad \xi = x - ct \quad \Longrightarrow \eta_x = 1,\, \eta_t = c, \quad \xi_x = 1,\, \xi_t = -c$$

Check the change of variables

$$\begin{aligned}
\partial_{tt}u - c^2\partial_{xx}u =& (\xi_t^2\partial_{\xi\xi}v + 2\xi_t\eta_t\partial_{\xi\eta}v + \eta_t^2\partial_{\eta\eta}v) - c^2(\xi_x^2\partial_{\xi\xi}v + 2\xi_x\eta_x\partial_{\xi\eta}v + \eta_x^2\partial_{\eta\eta}v) \\
=& (\xi_t^2 - c^2\xi_x^2)\partial_{\xi\xi}v + 2(\xi_t\eta_t - c^2\xi_x\eta_x)\partial_{\xi\eta}v + (\eta_t^2 - c^2\eta_x^2)\partial_{\eta\eta}v \\
=& 0 \cdot \partial_{\xi\xi}v + 2(-c^2 - c^2)\partial_{\xi\eta}v + 0 \cdot \partial_{\eta\eta}v = -4c^2\partial_{\xi\eta}v = 0
\end{aligned}$$

This is the canonical form of a hyperbolic PDE.

Hyperbolic canonical form solution

$$\partial_{\xi\eta} v = 0.$$

The general solution is given integrating in ξ and then in η, i.e.,

$$v(\xi, \eta) = \int^{\xi} \int^{\eta} \partial_{wz} v(w, z)\, dz\, dw = \int^{\xi} f(w) dw = F(\xi) + G(\eta),$$

where $\partial_{\xi} F(\xi) = f(\xi)$, for arbitrary functions f, G.
Recall the change of variables

$$\eta = x + ct, \quad \xi = x - ct.$$

So, the general solution of the wave equation is

$$u(x, t) = F(x - ct) + G(x + ct).$$

Hyperbolic canonical form for general hyperbolic PDE

Consider just the second order terms of the hyperbolic PDE $\Delta = B^2 - 4AC > 0$.

$$A\partial_{xx}u + Bu_{xy} + Cu_{yy} = 0.$$

We look for a change of variables $(x, y) \mapsto (\xi, \eta)$ such that the PDE simplifies to

$$\partial_{\xi\eta}v = 0.$$

The transformation can be applied noting that

$$\begin{aligned}
u_{xx} &= v_{\xi\xi}(\xi_x)^2 + 2v_{\xi\eta}\xi_x\eta_x + v_{\eta\eta}(\eta_x)^2, \\
u_{xy} &= v_{\xi\xi}\xi_x\xi_y + v_{\xi\eta}(\xi_x\eta_y + \xi_y\eta_x) + v_{\eta\eta}\eta_x\eta_y, \\
u_{yy} &= v_{\xi\xi}(\xi_y)^2 + 2v_{\xi\eta}\xi_y\eta_y + v_{\eta\eta}(\eta_y)^2,
\end{aligned}$$

The transformed PDE reads

$$(A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2)v_{\xi\xi} + (2A\xi_x\eta_x + B(\xi_x\eta_y + \xi_y\eta_x) + 2C\xi_y\eta_y)v_{\xi\eta} + (A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2)v_{\eta\eta} = 0.$$

$$(A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2)v_{\xi\xi} + (2A\xi_x\eta_x + B(\xi_x\eta_y + \xi_y\eta_x) + 2C\xi_y\eta_y)v_{\xi\eta} + (A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2)v_{\eta\eta} = 0.$$

We want to find the change of variables such that

$$\begin{cases} A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2 = 0 \\ A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2 = 0 \end{cases}$$

These are first order PDE, so we are looking for characteristics curves such that $\xi(x, y) = \text{const}$, if we find a curve, for example $y(x)$ such that $\xi(x, y(x)) = \text{const}$, then

$$\frac{d\xi}{dx} = \frac{\partial\xi}{\partial x} + \frac{\partial\xi}{\partial y}\frac{dy}{dx} = 0 \Longrightarrow \frac{dy}{dx} = -\frac{\partial_x\xi}{\partial_y\xi}.$$

From the first PDE, we then get

$$A\left(\frac{dy}{dx}\right)^2 - B\frac{dy}{dx} + C = 0,$$

which is called the characteristic equation for the original PDE. This is quadratic equation in $\frac{dy}{dx}$ with $\Delta = B^2 - 4AC > 0$. The two distinct solutions are

$$\frac{dy}{dx} = \frac{+B \pm \sqrt{\Delta}}{2A}.$$

From this we can get the transformation $(x, y) \mapsto (\xi, \eta)$ as we did in the linear PDE.

Example

$$u_{tt} + u_{tx} = 0$$

$$(\xi_t^2 + \xi_t \xi_x) u_{\xi\xi} + (2\xi_t \eta_t + \xi_t \eta_x + \xi_x \eta_t) u_{\xi\eta} + (\eta_t^2 + \eta_t \eta_x) u_{\eta\eta} = 0$$

The equations for ξ and η are the same equations.
We look for a curve $y(x)$ such that $\xi(x, y(x)) = \mathrm{const}$, i.e., $\xi(x, y(x)) = x + y(x) = \mathrm{const}$ and that

$$\begin{aligned}
&\xi_t^2 + \xi_t \xi_x = 0 \\
&\frac{\xi_t^2}{\xi_x^2} + \frac{\xi_t}{\xi_x} = 0 \\
&\left(\frac{dx}{dt}\right)^2 - \frac{dx}{dt} = 0 \\
&\frac{dx}{dt} = \begin{cases} 0 \\ 1 \end{cases} \Longrightarrow x(t) = \begin{cases} \xi_0 \\ \eta_0 + t \end{cases} \\
\Longrightarrow &\eta = x - t, \quad \xi = x.
\end{aligned}$$

Homework

1. Find the change of variables that transforms $u_{tt} + u_{tx} - 2u_{xx} = 0$ into the canonical form $\partial_{\xi\eta} v = 0$.
2. Solutions of wave equations $F(x - ct) + G(x + ct)$ are extremely similar to transport equation solutions $f(x - ct)$. Write the wave equations $u_{tt} - c^2 u_{xx} = 0$ as a system of two first order linear transport PDEs. (Hint introduce a new variable v such that $v_t = c^2 u_x$)*

$$(A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2)v_{\xi\xi} + (2A\xi_x\eta_x + B(\xi_x\eta_y + \xi_y\eta_x) + 2C\xi_y\eta_y)v_{\xi\eta} + (A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2)v_{\eta\eta} = 0.$$

What if we try to do the same with a parabolic PDE $\Delta = 0$?

$$\frac{dx}{dt} = \frac{B \pm \sqrt{\Delta}}{2A} = \frac{B}{2A}.$$

There is only one characteristic curve. So, choosing $\eta = 2Ay - Bx$ and $\xi = x$ (or anything else lin. independent from η), we have that $C' = 0$, which means that also $B' = 0$ (as $\Delta = B^2 - 4AC = 0$), so we get the canonical form

$$A'\frac{\partial^2 v}{\partial \xi^2} = 0,$$

with the general solution $v(\xi, \eta) = F(\eta) + \xi G(\eta) = F(2Ax - By) + xG(2Ax - By)$.

$$(A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2)v_{\xi\xi} + (2A\xi_x\eta_x + B(\xi_x\eta_y + \xi_y\eta_x) + 2C\xi_y\eta_y)v_{\xi\eta} + (A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2)v_{\eta\eta} = 0.$$

What if we try to do the same with an elliptic PDE?

There is no characteristics that is conserved. But, one can instead eliminate the coefficient of $u_{\xi\eta}$ to obtain the canonical form for the elliptic PDE. Using $\eta = x$ and $\xi = \frac{2Ay - Bx}{\sqrt{-\Delta}}$, we get

$$A\left(\frac{\partial^2 v}{\partial \xi^2} + \frac{\partial^2 v}{\partial \eta^2}\right) = 0.$$

We cannot rewrite it as an ODE. One could use complex functions to solve it.

Different approach: Fourier space

Suppose that we consider only periodic functions on $[-\pi, \pi]^2$, we can they something on the solutions. Since, we are periodic and boundaries play a marginal role, we better consider a right hand side g that is periodic and we look for periodic solutions. We can then use Fourier series to solve the PDE, which is a powerful tool for linear PDEs with constant coefficients.

Exact solutions for periodic elliptic equations (Fourier) (1/3)

Eigenfunctions of the differential operator

First of all, let's notice that the trigonometric functions are special functions for the differential operator

$$\begin{aligned}
\partial_x e^{ixk} &= ike^{ixk}, & \partial_{xx} e^{ixk} &= -k^2 e^{ixk}, \\
\partial_x \sin(kx) &= k\cos(kx), & \partial_{xx} \sin(kx) &= -k^2 \sin(kx), \\
\partial_x \cos(kx) &= -k\sin(kx), & \partial_{xx} \cos(kx) &= -k^2 \cos(kx).
\end{aligned}$$

Recall:

$$e^{i\pi} = -1, \qquad \sin(x) = \frac{e^{ix} - e^{-ix}}{2i}, \qquad \cos(x) = \frac{e^{ix} + e^{-ix}}{2},$$

$$e^{ix} = \cos(x) + i\sin(x), \qquad e^{-ix} = \cos(x) - i\sin(x).$$

So we focus on the trigonometric functions of the type e^{ixk}.

Exact solutions for periodic elliptic equations (Fourier) (2/3)

Fourier series

For simplicity let's consider $\Omega = [-\pi, \pi]$ with periodic boundary conditions. We can decompose any periodic function g in Fourier series if $g \in L^2(\Omega)$.

$$g(x) = \sum_{k \in \mathbb{Z}} \hat{g}_k e^{ikx}, \qquad \hat{g}_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(x) e^{-ikx} \mathrm{d}x.$$

Parseval theorem

$$\|\hat{\mathbf{g}}\|_2^2 = \sum_{k \in \mathbb{Z}} |\hat{g}_k|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |g(x)|^2 \mathrm{d}x = \frac{1}{2\pi} \|g\|_2^2.$$

Wikipedia page on Fourier series
Youtube playlist of 3Blue1Brown on Fourier series
Youtube video on solving heat equations with Fourier

Exact solutions for periodic elliptic equations (Fourier) (3/3)

$$-u_{xx} - u_{yy} = g$$

Exploiting linearity for heat equation

Let's us use the ansatz $u(x,y) = \sum_{k\in\mathbb{Z}} \sum_{j\in\mathbb{Z}} \hat{u}_{k,j} e^{ikx} e^{ijy}$, where $\hat{u}_{k,j}$ are the Fourier coefficients of the solution.
The elliptic PDE with a periodic source term $g(x,y) = \sum_{k\in\mathbb{Z}} \sum_{j\in\mathbb{Z}} \hat{g}_{k,j} e^{ikx} e^{ijy}$ with $\hat{g}_{0,0} = 0$ reads

$$-(u_{xx}(x,y) + u_{yy}(x,y)) = g(x,y)$$

We can rewrite it as

$$-\sum_{k\in\mathbb{Z}} \sum_{j\in\mathbb{Z}} \hat{u}_{k,j}(-k^2 - j^2) e^{ikx} e^{ijy} = \sum_{k\in\mathbb{Z}} \sum_{j\in\mathbb{Z}} \hat{g}_{k,j} e^{ikx} e^{ijy}$$

$$\Longrightarrow \hat{u}_{k,j} = \frac{\hat{g}_{k,j}}{k^2 + j^2} \quad \text{for } (k,j) \neq (0,0), \qquad \hat{u}_{0,0} \text{ is arbitrary.}$$

The last step is possible because the trigonometric functions are orthogonal, so we can directly compare only the coefficients of the two sides of the equality (see next slide).

Orthogonality of trigonometric functions

$$\int_{-\pi}^{\pi} e^{ikx} e^{-ijx} \mathrm{d}x = \int_{-\pi}^{\pi} e^{i(k-j)x} \mathrm{d}x = \begin{cases} 2\pi, & k = j, \\ 0, & k \neq j, \end{cases} = \delta_{kj} 2\pi.$$

Indeed, for any $z \in \mathbb{Z}$ with $z \neq 0$, we have

$$\int_{-\pi}^{\pi} e^{izx} \mathrm{d}x = \int_{-\pi}^{\pi} \cos(zx) \mathrm{d}x + i \int_{-\pi}^{\pi} \sin(zx) \mathrm{d}x = \left[\frac{\sin(zx)}{z} \right]_{-\pi}^{\pi} + i \left[-\frac{\cos(zx)}{z} \right]_{-\pi}^{\pi} = 0, \text{ or}$$

$$\int_{-\pi}^{\pi} e^{izx} \mathrm{d}x = \left[\frac{e^{izx}}{iz} \right]_{-\pi}^{\pi} = \frac{e^{iz\pi} - e^{-iz\pi}}{iz} = \begin{cases} \frac{1-1}{iz} = 0, & z \text{ even}, \\ \frac{-1-(-1)}{iz} = 0, & z \text{ odd}. \end{cases}$$

Hence, if we know that $\sum_{k \in \mathbb{Z}} a_k e^{ikx} = \sum_{k \in \mathbb{Z}} b_k e^{ikx}$, then we can multiply both sides by e^{-ijx}, for any $j \in \mathbb{Z}$ and integrate to get

$$\begin{aligned} \int_{-\pi}^{\pi} \sum_{k \in \mathbb{Z}} a_k e^{ikx} e^{-ijx} \mathrm{d}x &= \int_{-\pi}^{\pi} \sum_{k \in \mathbb{Z}} b_k e^{ikx} e^{-ijx} \mathrm{d}x \\ \implies \sum_{k \in \mathbb{Z}} a_k \int_{-\pi}^{\pi} e^{ikx} e^{-ijx} \mathrm{d}x &= \sum_{k \in \mathbb{Z}} b_k \int_{-\pi}^{\pi} e^{ikx} e^{-ijx} \mathrm{d}x \\ \implies \sum_{k \in \mathbb{Z}} a_k \delta_{kj} 2\pi &= \sum_{k \in \mathbb{Z}} b_k \delta_{kj} 2\pi \implies a_j = b_j. \end{aligned}$$

Existence, uniqueness and well-posedness

For the PDEs above we have found classes of solutions. How can we find unique solutions to specific problems? What should we need to specify?

Definition (Cauchy problem)

Consider a PDE of order k in $\Omega \subset \mathbb{R}^d$ and let S be a given smooth surface on $\mathbb{R}^d$. Let also $n = n(x)$ denote the unit normal vector to the surface S at a point $x = (x_1, x_2, \ldots, x_d) \in S$. Suppose that on any point x of the surface S the values of the solution u and of all its directional derivatives up to order $k-1$ in the direction of n are given, i.e., we are given functions $f_0, f_1, \ldots, f_{k-1} : S \to \mathbb{R}$ such that

$$u(x) = f_0(x), \text{ and } \frac{\partial u}{\partial n}(x) = f_1(x), \text{ and } \frac{\partial^2 u}{\partial n^2}(x) = f_2(x), \ldots, \text{ and } \frac{\partial^{k-1} u}{\partial n^{k-1}}(x) = f_{k-1}(x).$$

The **Cauchy problem** consists of finding the unknown function(s) u that satisfy simultaneously the PDE and the conditions above, which are called the **initial conditions** (ICs) and the given functions $f_0, f_1, \ldots, f_{k-1}$, will be referred to as the initial data.
According to the role of the ICs they can be called also **boundary conditions** (BCs).

Examples (Cauchy problem for transport equation)

$$\begin{cases} u_t + u_x = 0, & (t,x) \in \Omega = \mathbb{R}^+ \times \mathbb{R}, \\ u(0,x) = \sin(x), & x \in \mathbb{R}. \end{cases}$$

Here, $S = \{(t,x) \in \mathbb{R}^2 : t = 0\}$.
The general solution of the transport equation is $u(x,t) = f(x-t)$, so that the initial condition reads $f(x) = \sin(x)$, i.e., $u(x,t) = \sin(x-t)$.

Examples (Cauchy problem for wave equation)

$$\begin{cases} u_{tt} - u_{xx} = 0, & (t,x) \in \Omega = \mathbb{R}^+ \times \mathbb{R}, \\ u(t,0) = \sin(t), & t \in \mathbb{R}^+, \\ u_x(t,0) = 0, & t \in \mathbb{R}^+. \end{cases}$$

In this case, $S = \{(t,x) \in \mathbb{R}^2 : x = 0\}$ and $n = (n_t, n_x) = (0,1)$. The general solution of the wave equation is $u(t,x) = f(x-t) + g(x+t)$, so that the initial (boundary) conditions read $f_0(t) = \sin(t)$ and $f_1(t) = 0$, so

$$\begin{cases} f(-t) + g(t) = \sin(t), \\ f'(-t) + g'(t) = 0, \end{cases} \Longrightarrow f(\xi) = \frac{1}{2}\sin(-\xi) + C, \quad g(\eta) = \frac{1}{2}\sin(\eta) - C,$$

so, $u(t,x) = \frac{1}{2}(\sin(x+t) + \sin(-x+t))$.

Theorem (Cauchy-Kovalesvskaya Theorem)

Consider a Cauchy problem for a linear PDE, let x^0 be a point of the initial surface S, which is assumed to be *analytic* (very regular). Suppose that S is not a *characteristic surface* at the point x^0. Assume that all the coefficients of the linear PDE, the right-hand side g, and all the initial data $f_0, f_1, \ldots, f_{k-1}$ are *analytic* functions on a neighbourhood of the point x^0. Then, the Cauchy problem **has a solution** u, defined in the neighbourhood of x^0. Moreover, the solution u is *analytic* in a neighbourhood of x^0 and it is **unique** in the class of analytic functions.

- Assumptions: Regularity
- Assumptions: Linearity of PDE
- Outcome: Existence
- Outcome: Uniqueness
- Outcome: Regularity of the solution

- Is this enough? No, the solution might still mis-behave

Well-posedness

Definition

A PDE problem is well-posed if:

1. The PDE has a solution
2. The solution is unique
3. The solution depends continuously on the PDE coefficients and on the problem data (IC/BC)

If the PDE problem is not well-posed, we say it is ill-posed.

Exercise

Show that the solution of the Cauchy problem for the wave equation

$$\begin{cases} \partial_{tt} u - \partial_{xx} u = 0, \\ u(t, 0) = f(t), \\ u_x(t, 0) = g(t) \end{cases}$$

for some known BCs f and g is given by the d'Alembert's formula

$$u(t, x) = \frac{1}{2}(f(t - x) + f(t + x)) + \frac{1}{2} \int_{t-x}^{t+x} g(s) \mathrm{d}s.$$

Show that the Cauchy problem is well-posed (skipping the uniqueness). *

Exercise (Difficult) Dubrovin's notes

1. Find the solution of the Laplace equation on the periodic domain $\Omega = [0, 2\pi]$ for various k

$$\begin{cases} \partial_{tt} u + \partial_{xx} u = 0, \\ u(0, x) = 0, \\ u_t(0, x) = \frac{\sin(kx)}{k}, \\ u(t, 0) = u(t, 2\pi). \end{cases}$$

Steps:

- $u_k = \frac{a_0(t)}{2} + \sum_{n=1}^{\infty} [a_n(t) \cos(nx) + b_n(t) \sin(nx)]$
- Substitute in the equation and find the general solution using the method of separation of variables *(different setting from the Fourier series we have sen before)*
- $\partial_{tt} a_n(t) = n^2 a_n(t)$ for all n with $a_n(0) = 0, \partial_t a_n(0) = 0$
- $\partial_{tt} b_n(t) = n^2 b_n(t)$ for all n with $b_n(0) = 0, \partial_t b_n(0) = 0$ for $n \neq k$, $\partial_t b_k(0) = 1/k$.
- $u_k(t, x) = \frac{1}{k^2} \sin(kx) \sinh(kt)$

2. Even if $\sup_x |u_k(0, x)| + |\partial_t u_k(0, x)|$ is small, we can find large enough k so that for any time $t_0 > 0$ $\sup_x |u_k(t_0, x)| + |\partial_t u_k(t_0, x)|$ is large.

Theorem *

Let $u_k(t,x) = \frac{1}{k^2}\sin(kx)\sinh(kt)$. For any positive ε, M, t_0 there exists an integer K such that for any $k > K$ the initial data satisfies $\sup_x |u_k(0,x)| + |\partial_t u_k(0,x)| < \varepsilon$ but the solution at the time t_0 satisfies $\sup_x |u_k(t_0,x)| + |\partial_t u_k(t_0,x)| > M$.

Proof: Choosing an integer K_1 satisfying $K_1 > \frac{1}{\epsilon}$ we will have the initial condition inequality for any $k \geq K_1$. In order to obtain a lower estimate of the second form at time t_0 let us first observe that

$$\sup_{x\in[0,2\pi]} (|u_k(x,t)| + |\partial_t u_k(x,t)|) = \frac{1}{k^2}\sinh(kt) + \frac{1}{k}\cosh(kt) > \frac{1}{k^2}e^{kt}$$

where we have used an obvious inequality $\frac{1}{k} > \frac{1}{k^2}$ for $k > 1$.

The function $y = \frac{e^x}{x^2}$ is monotone increasing for $x > 2$ and $\lim_{x\to+\infty} \frac{e^x}{x^2} = +\infty$.

Hence for any $t_0 > 0$ there exists x_0 such that $\frac{e^x}{x^2} > \frac{M}{t_0^2}$ for $x > x_0$.

Let K_2 be a positive integer satisfying $K_2 > \frac{x_0}{t_0}$.

Then for any $k > K_2$

$$\frac{e^{kt_0}}{k^2} = t_0^2\frac{e^{kt_0}}{k^2t_0^2} > t_0^2\frac{e^{x_0}}{x_0^2} > M.$$

Choosing $K = \max(K_1, K_2)$ we complete the proof of the Theorem.

Take home message

Not all boundary conditions are suitable for having a well-posed problem even if we are looking at constant coefficient linear PDEs.